

A flexible novel extension of discrete generalized Pareto distribution

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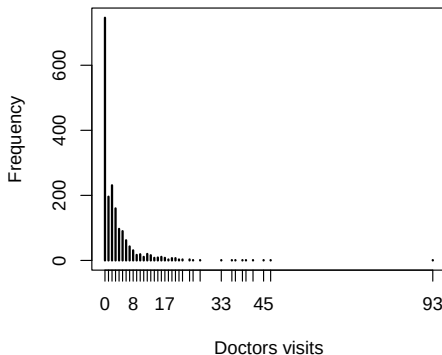
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- ① Motivation
- ② Introduction & background
- ③ Flexible model construction
- ④ Applications
- ⑤ Future research

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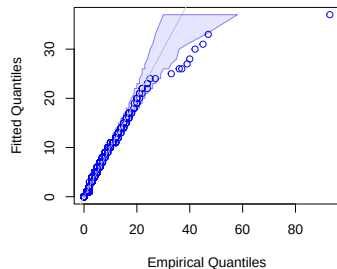
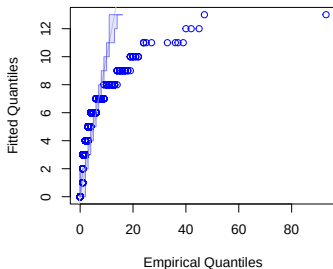
Motivation

- We consider the doctors visits data set from zic R package. The data has 746 zero observations.



Fitted existing models

- We fitted zero-inflated negative binomial and zero-inflated Poisson distributions.



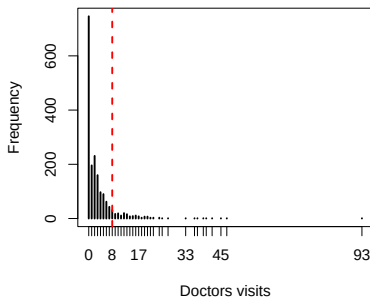
- We required a model that correctly captures the lower and upper tail behavior.

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Discrete generalized Pareto distribution

- On possibility is to model threshold exceedances using **discrete generalized Pareto distribution** (DGPD) Hitz et al. (2017). The DGPD is defined by discretizing $\mathcal{F}(k) = 1 - (1 + k\xi/\beta)^{-1/\xi}$

$$\Pr(Y - t = k | Y \geq t) = \mathcal{F}(k+1; \beta, \xi) - \mathcal{F}(k; \beta, \xi), \quad k \in \mathbb{N}_0 \quad (1)$$



Problems with DGPD fitting

Problem with DGPD fitting...

- Optimal threshold selection is problematic.
- Losing a large amount of data when using high quantile for threshold selection.
- We required a **flexible model** that accurately models count data across different scenarios:
 - ① the entire data with a heavy tail
 - ② the data with excessive zeros and heavier tail together
 - ③ the data above a very low threshold

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Discrete extended generalized Pareto distribution (DEGDP)

- Construction is based on the integral transform idea to simulate GPD random draws Naveau et al. (2016), that is

$$F_{\beta,\xi}^{-1}(U), \quad U \sim \mathcal{U}(0, 1)$$

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where $\mathcal{G}$ is a CDF on $[0, 1]$ and $U \sim \mathcal{U}(0, 1)$.

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- The **key problem** is to find a class for $\mathcal{G}$ which preserves the upper tail behavior with shape parameter ξ and also controls the lower tail behavior

DEGPD

- Naveau et al. (2016) defined restrictions for validity of G families. For instance, the tail of $\mathcal{G}$ denoted by $\bar{\mathcal{G}} = 1 - \mathcal{G}$ has to satisfy

$$\lim_{u \rightarrow 0} \frac{\bar{\mathcal{G}}(1-u)}{u} = a, \text{ for some finite } a > 0 \text{ (upper tail behavior),}$$
$$\lim_{u \rightarrow 0} \frac{\mathcal{G}(u)}{u^\kappa} = c, \text{ for some finite } c > 0 \text{ (lower tail behavior).} \quad (3)$$

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- We start by considering the CDF $\mathcal{G}\{F(.;\beta,\xi)\}$ of EGPD where $\mathcal{G}$ meets the conditions (3). To model non-negative integer data, we discretize the CDF by

$$\Pr(Y = k) = \mathcal{G}\{\mathcal{F}(k+1;\beta,\xi)\} - \mathcal{G}\{\mathcal{F}(k;\beta,\xi)\}, \quad k \in \mathbb{N}_0 \quad (4)$$

The distribution will be called **DEGPD**.

Zero-inflated DEGPD

Zero inflation problem along with heavier tail...

Let Z be observed with an **excessive number of zeros** relative to those observed under the DEGPD. The zero-inflated distribution (**ZIDEGPD**) is defined in a straightforward way as:

$$\Pr(Z = m) = \begin{cases} \pi + (1 - \pi)\mathcal{G}\{\mathcal{F}(1, \beta, \xi)\} & m = 0 \\ (1 - \pi)[\mathcal{G}\{\mathcal{F}(m + 1, \beta, \xi)\} - \mathcal{G}\{\mathcal{F}(m, \beta, \xi)\}] & m \in \mathbb{N} \end{cases} \quad (5)$$

where $0 \leq \pi \leq 1$ and the remaining parameters would be the same as DEGPD.

It turns out that the CDF of ZIDEGPD is

$$\Pr(Z \leq m) = \pi + (1 - \pi)\mathcal{G}\{\mathcal{F}(m + 1, \beta, \xi)\}, \quad m \in \mathbb{N}_0 \quad (6)$$

$\mathcal{G}$ families

- For $\mathcal{G}$, we use different parametric distributions that preserve upper tail behavior (Gamet and Jalbert, 2022).

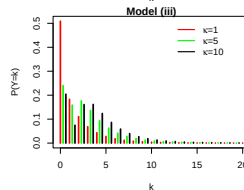
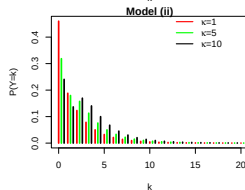
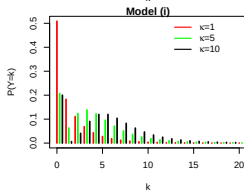
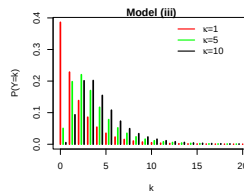
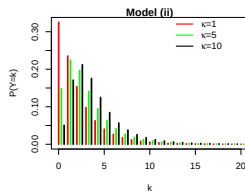
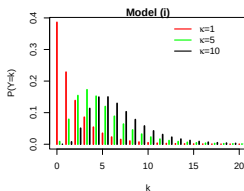
(i) (M_1): $\mathcal{G}(u; \kappa) = u^\kappa, \kappa > 0;$

(ii) (M_2): $\mathcal{G}(u; \kappa) = \frac{2}{2\Phi(\sqrt{\kappa})-1} \left[\Phi \{ \sqrt{\kappa}(u-1) \} - \{1 - \Phi(\sqrt{\kappa})\} \right]$

iii) (M_3): $\mathcal{G}(u; \kappa) = \begin{cases} 0 & \text{for } u < 0; \\ \frac{D_{\{(1/2-\omega)u-\omega\}}(\kappa, \kappa) - D_\omega(\kappa, \kappa)}{D_{1/2}(\kappa, \kappa) - D_\omega(\kappa, \kappa)} & \text{for } 0 \leq u \leq 1; \\ 1 & \text{for } u > 1; \end{cases}$

We choose $\omega = 1/32$ here.

PMF behaviors of DEGPD and ZIDEGPD



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Estimation

- The likelihood function for DEGPD (4) and ZIDEGPD (5) is written as.

$$l(\boldsymbol{\theta}) = \sum_{i=1}^n \log [\mathcal{G}(\mathcal{F}(y_i + 1; \beta, \xi); \kappa) - \mathcal{G}(\mathcal{F}(y_i; \beta, \xi); \kappa)]. \quad (7)$$

and

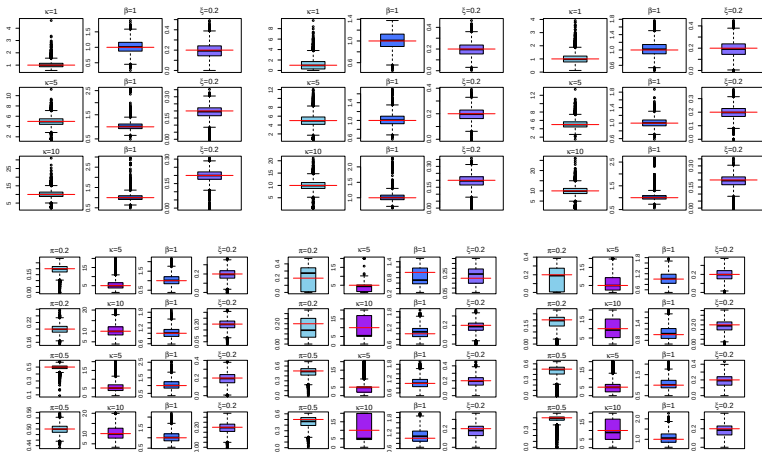
$$\begin{aligned} l(\boldsymbol{\theta}) = & \sum_{i=1}^n l_0(y_i) \log [\pi + (1 - \pi) \mathcal{G}(\mathcal{F}(1; \beta, \xi); \kappa)] \\ & + \sum_{i=1}^n (1 - l_0(y_i)) \log(1 - \pi) \times \left[\mathcal{G}(\mathcal{F}(y_i + 1; \beta, \xi); \kappa) \right. \\ & \left. - \mathcal{G}(\mathcal{F}(y_i; \beta, \xi); \kappa) \right]. \end{aligned} \quad (8)$$

Where $\boldsymbol{\theta}$ is a vector of unknown parameters.

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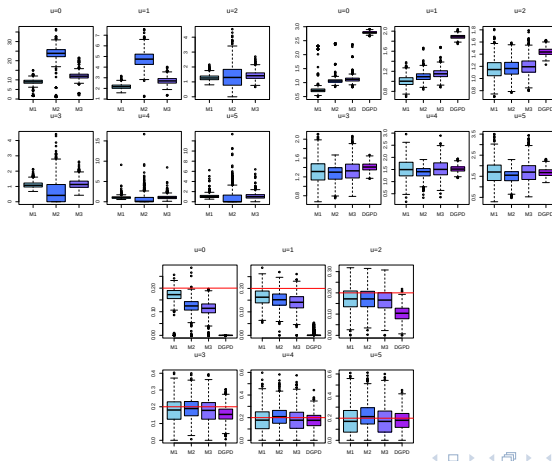
Simulation study (data with zero inflation)

- The sample size $n = 1000$ with 10^4 replications used to estimate the parameters



Simulation study (proceeded with low threshold)

We simulated heavy-tailed (10^3) samples of size $n = 2000$ from $DGEV(\mu = 2, \beta = 1, \xi = 1/20)$.

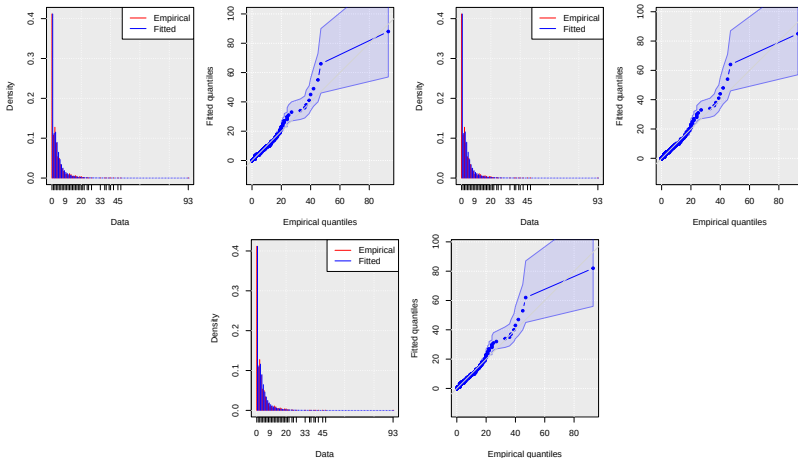


Application to doctors visit data set

- We apply ZIDEPGD to doctors visit data set. The data set has 746 zero values among 1812.

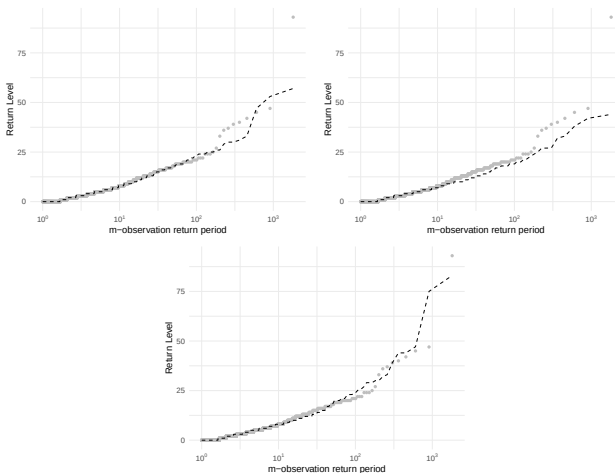
Doctors visits					
	π	κ	β	ξ	BIC
M_1	0.38 [0.32, 0.41]	4.34 [1.87, 12.36]	1.23 [0.56, 2.41]	0.40 [0.26, 0.50]	10242.039
M_2	0.37 [0.31, 0.41]	9.69 [3.56, 20.00]	1.67 [1.17, 2.52]	0.38 [0.26, 0.47]	10241.567
M_3	0.38 [0.33, 0.41]	5.28 [2.25, 10.55]	1.77 [1.22, 1.55]	0.37 [0.25, 0.47]	10240.84

Real application



Real application

Return levels corresponding to m-observation return periods



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Future research

- New flexible EGPD and DEGPD distribution can be proposed using any parametric family for $\mathcal{G}$, which fulfills the condition discussed earlier.
- The proposed models can be used in a distributional regression modeling context (Ahmad et al., 2022).
- Space-time modeling of discrete extreme events
- We developed an R code for fitting these models, and it will be available soon on [GitHub](https://github.com/touqeerahmadunipd)¹

¹<https://github.com/touqeerahmadunipd>.

PhD scholarships

- I encourage all master's students to look for scholarships actively and put applications with well-written, motivated research proposals. For scholarship updates, you can join the WhatsApp group. Or

PhD Statistics Aspirants

WhatsApp group



- You can write me an email at touqeer.ahmad@ensai.fr.

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- Gamet, P. and Jalbert, J. (2022). A flexible extended generalized pareto distribution for tail estimation. *Environmetrics*, 33(6):e2744.
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- Naveau, P., Huser, R., Ribereau, P., and Hannart, A. (2016). Modeling jointly low, moderate, and heavy rainfall intensities without a threshold selection. *Water Resources Research*, 52(4):2753–2769.

Thank you for your attention!